

Introduction to CSS

CSS makes websites beautiful and attractive. Today we move from structure to style — transforming plain HTML pages into visually polished, professional web experiences.

1

What is CSS?

2

Types of CSS

3

Selectors

4

Colors & Backgrounds

5

Borders, Margin & Padding

6

Width & Height

7

Fonts & Text

8

Practical Task



Today's Goal: Style your previous HTML pages using the CSS techniques we learn today.

What is CSS?

CORE CONCEPT

CSS stands for **Cascading Style Sheets**. It is the language used to style and visually format HTML web pages. While HTML provides the structure and content, CSS controls how that content looks — including colors, fonts, spacing, layout, and more.

```
/* Example CSS */  
h1 {  
  color: blue;  
}
```

This simple rule turns every `<h1>` heading blue on your page.

CSS Controls

- Colors
- Layout
- Fonts
- Spacing
- Borders & Alignment

Without CSS

Plain, unformatted, hard to read

With CSS

Colorful, professional, engaging

Types of CSS

3 METHODS

1

Inline CSS

CSS written directly inside an HTML tag using the `style` attribute. Quick but hard to maintain across multiple elements.

```
<h1 style="color:red;">
  Welcome
</h1>
```

2

Internal CSS

CSS written inside a `<style>` block in the `<head>` section of your HTML file. Good for single-page styling.

```
<style>
  h1 { color: blue; }
</style>
```

3

External CSS

CSS written in a separate `.css` file and linked to the HTML. This is the professional, industry-standard approach for real projects.

```
<link rel="stylesheet"
      href="style.css">
```

✓ ★ **Best Practice:** External CSS is the preferred method in real-world projects — it keeps your code clean, organized, and reusable across multiple pages.

CSS Selectors

Selectors are the backbone of CSS — they tell the browser **which HTML elements to style**. Understanding the three core selector types gives you precise control over your entire page.



Element Selector

Targets all elements of a given type.

```
p {  
  color: red;  
}
```



Class Selector

Targets elements with a specific class.
Reusable on multiple elements.

```
.box {  
  background-color: yellow;  
}
```



ID Selector

Targets a single unique element by its ID. Use only once per page.

```
#title {  
  color: blue;  
}
```

Example HTML

```
<h1 id="title">Hello</h1>  
<p class="box">Content</p>  
<p>Another paragraph</p>
```



Tip: Classes start with `.` and IDs start with `#` in your CSS. Use classes when you want to reuse a style, and IDs for one unique element.

Colors and Backgrounds

Text & Background Color

```
h1 {  
  color: purple;  
}  
  
body {  
  background-color: lightblue;  
}
```

Background Image

```
body {  
  background-image:  
  url("bg.jpg");  
}
```

Common Color Formats

Color Names

red, blue, lightblue

HEX Codes

color: #ff0000;

RGB Values

color: rgb(255, 0, 0);

- ❏ HEX codes are the most commonly used format in professional web design.



Borders, Margin & Padding

CSS Code

```
/* Border */
div {
  border: 2px solid black;
}

/* Margin - space OUTSIDE */
div {
  margin: 20px;
}

/* Padding - space INSIDE */
div {
  padding: 15px;
}
```

Visual Box Model



Margin

Space **outside** the border — pushes elements away from each other



Border

The visible **outline** surrounding the element



Padding

Space **inside** the border — between the border and the content

  **Memory Trick:** Think of a picture frame — the padding is the mat inside the frame, the border is the frame itself, and the margin is the wall space around the frame.

Width and Height

The `width` and `height` properties let you control exactly how large an element appears on screen. These are essential for building cards, containers, image boxes, and page sections.

```
/* Width only */
```

```
div { width: 300px; }
```

```
/* Height only */
```

```
div { height: 200px; }
```

```
/* Combined example */
```

```
.box {  
  width: 250px;  
  height: 150px;  
  background-color: pink;  
}
```

Common Use Cases



Cards



Containers



Sections



Images



You can also use `%` or `vw/vh` units for responsive sizing that adapts to screen size.

Fonts and Text Styling

Typography is one of the most powerful tools in web design. CSS gives you fine-grained control over how text looks and feels on the page.

1

Font Size

```
p{  
  font-size: 20px;  
}
```

2

Font Family

```
p{  
  font-family: Arial;  
}
```

3

Font Weight

```
p{  
  font-weight: bold;  
}
```

4

Text Alignment

```
h1{  
  text-align: center;  
}
```

5

Text Decoration

```
a{  
  text-decoration: none;  
}
```


Complete CSS Example: Profile Card

HTML

```
<div class="card">
  <h1>Harinath</h1>
  <p>Software Architect</p>
</div>
```

CSS

```
.card {
  width: 300px;
  padding: 20px;
  background-color: lightyellow;
  border: 2px solid black;
  text-align: center;
}

h1 {
  color: blue;
}
```

What This Example Demonstrates

- **Width & Padding**
Controls size and inner spacing of the card
- **Background Color**
Gives the card a distinct visual appearance
- **Border**
Creates a visible outline around the card
- **Text Alignment & Color**
Centers content and applies heading color

✔️ This card combines all the CSS properties from today's lesson into one practical, real-world example.

Practical Task & Homework

In-Class Task: Style Previous HTML Pages

Open your existing HTML pages and apply CSS to improve their appearance. Focus on the following properties:

Colors

Fonts

Backgrounds

Borders

Margin

Padding

Pages to style: About Me Page · Student Registration Form ·
Timetable

Homework: Design a Colorful Profile Card

Build a styled profile card that showcases everything you've learned today. Your card must include:

- Your name and a profile image
- A list of skills and a short bio
- Social media links with `text-decoration: none`

  **Challenge:** Use background colors, borders, padding, and font styling together to make your card stand out.
"Good design improves user experience."